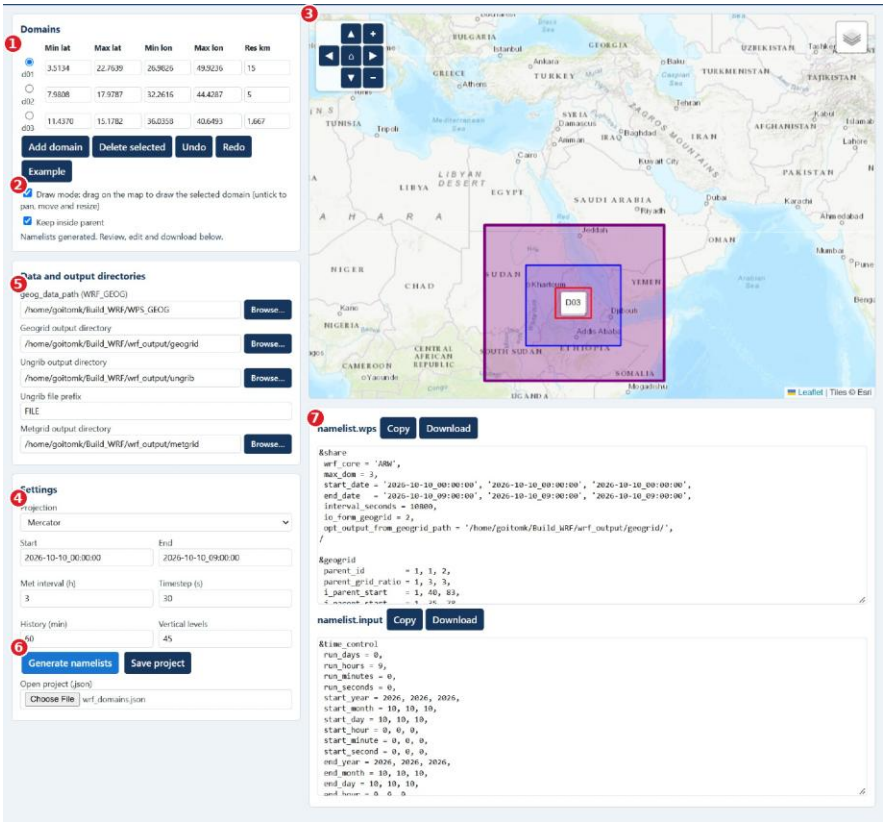
WRF Domain Wizard: User Guide

This guide walks through the web app from an empty map to namelist.wps and namelist.input. Red numbers on each picture match the numbered points beside it.



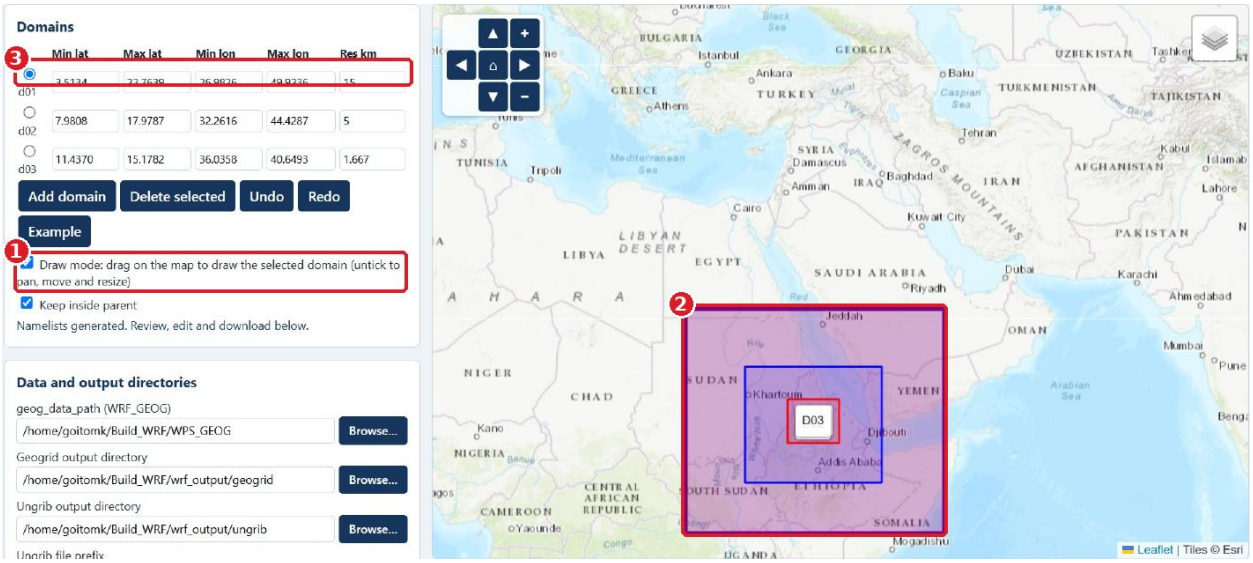
No.	Area
1	Domains table: extents and resolution of each domain
2	Draw mode and Keep inside parent options
3	Map with the navigation pad and layer switcher
4	Settings: projection and timing
5	Data and output directories
6	Generate namelists button
7	Generated namelist.wps and namelist.input, with Copy and Download

Before you start

- Open the live page (https://goikelem.github.io/WRF-domain-Wizard/) or open index.html in a browser. Chrome or Edge is recommended.
- An internet connection is needed for the map and tiles.
- Click Example at any time to load a 3-domain sample.

Step 1. Draw the outer domain (d01)

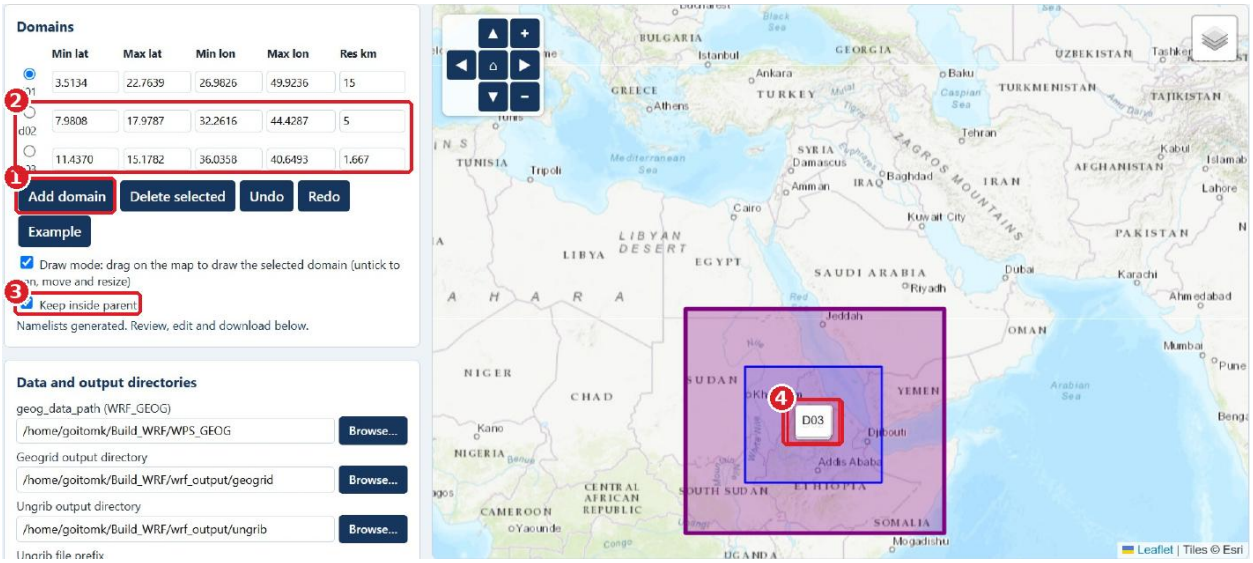
Draw mode is already on when the page opens, so you can start drawing straight away.



1. **Draw mode**: keep this box ticked. The cursor becomes a crosshair and dragging the map is switched off.
2. **Drag on the map**: press and hold at one corner of your area, move to the opposite corner and release. If you move the pointer close to the map edge while holding, the map scrolls by itself, so the far corner can be outside the current view.
3. **Domains table**: the d01 extents (Min/Max lat and lon) fill in automatically. You can also type them here, and set the resolution in km.

Step 2. Add nested domains (d02, d03, ...)

Each nest is drawn inside the previous one.

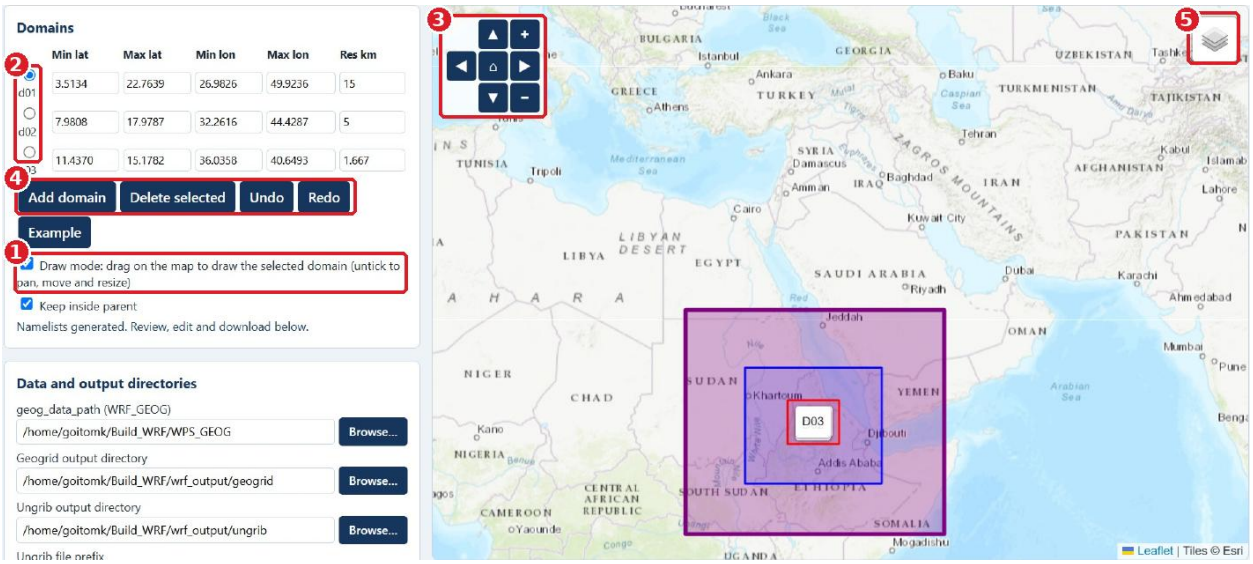


1. **Add domain**: adds a new row, up to 9 domains. The new resolution is one third of the previous one.
2. **New rows (d02, d03)**: change the resolution if you need another ratio. The extents stay empty until you draw or type them.

3. **Keep inside parent**: stops a nest from being drawn or moved outside its parent box.
4. **Draw the nest**: drag a box inside its parent. The box is labelled D02, D03 and so on.

Step 3. Select, move and resize boxes

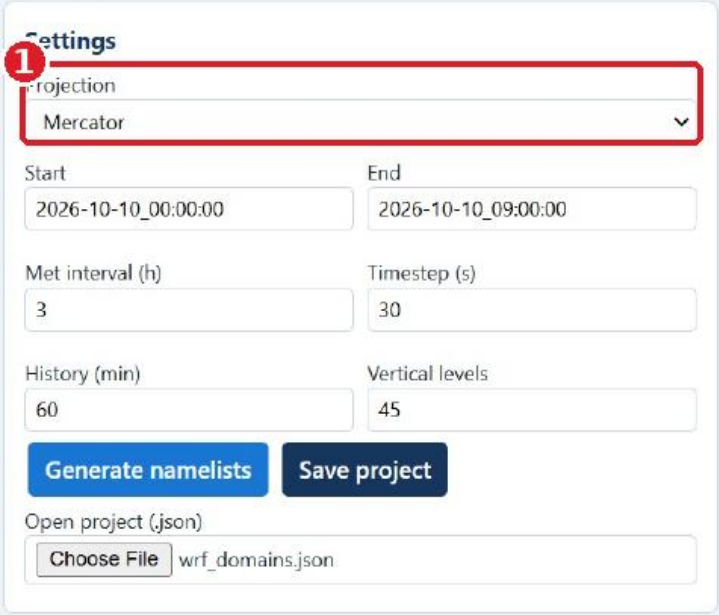
Untick Draw mode first. Then dragging works on the boxes and the map instead of drawing new ones.



1. **Draw mode**: untick it (it is still ticked in this screenshot). Dragging the map now pans it.
2. **Select a domain**: tick its radio button, or click the box on the map. The selected box gets a thicker outline and white squares.
3. **Navigation pad**: the arrows pan the map, + and − zoom, and the house button fits all domains. With a box selected, drag its centre square to move it and a corner square to resize it.
4. **Undo, Redo and Delete selected**: step back or forward, or remove the selected domain (later ones are renumbered).
5. **Layer switcher**: the map starts on ESRI World Topo. Choose ESRI World Imagery for a satellite view.

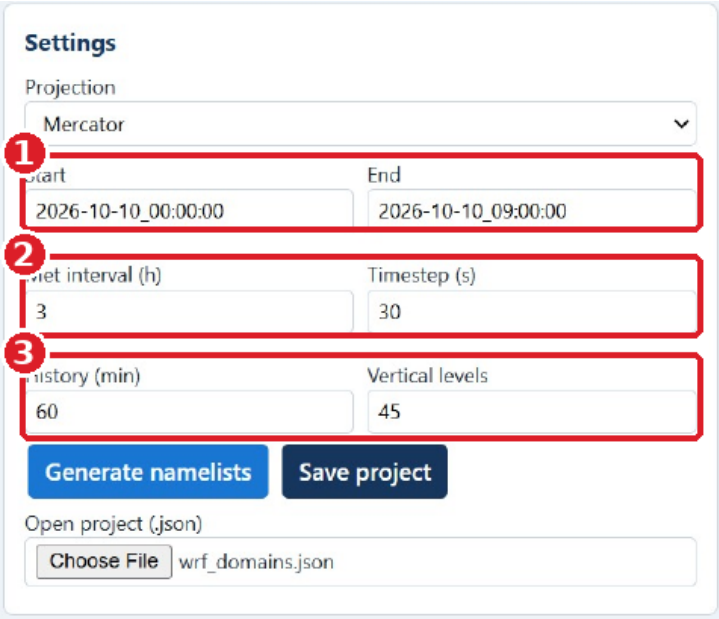
Step 4. Choose the projection

The reference point and true latitudes are calculated from d01.



1. **Projection**: choose Lambert Conformal, Mercator, Polar Stereographic or Latitude-Longitude. The screenshot uses Mercator.

Step 5. Set the timing and run options



1. **Start and End**: use the format YYYY-MM-DD_HH:MM:SS. End must be later than Start.
2. **Met interval (h) and Timestep (s)**: how often your input data arrives, and the WRF time step.
3. **History (min) and Vertical levels**: output interval and number of vertical levels (e_vert).

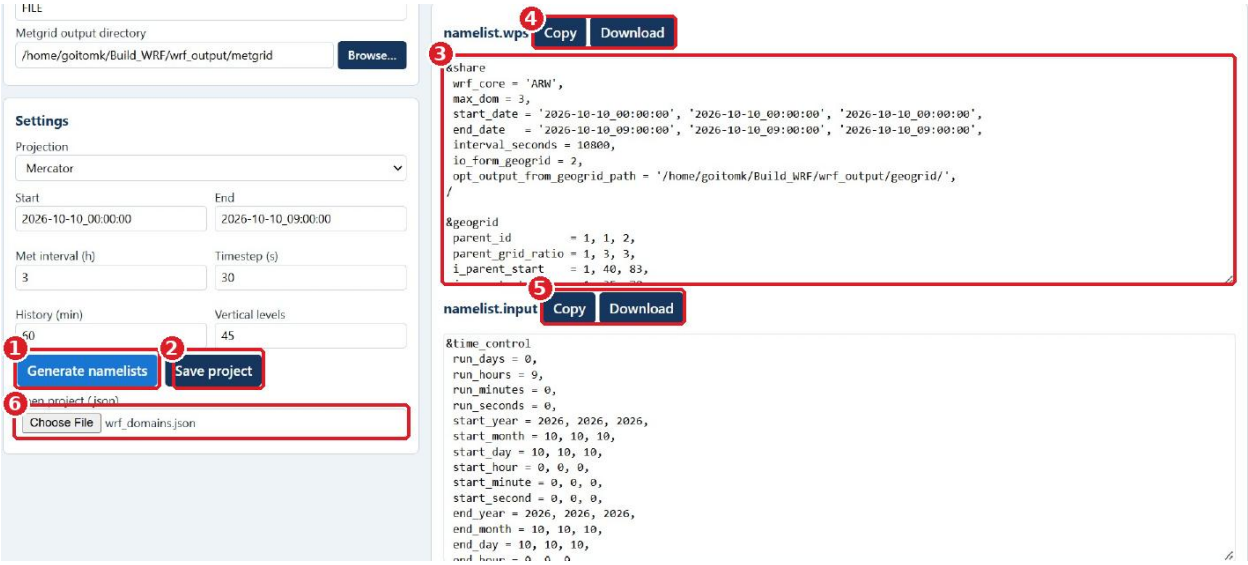
Step 6. Set the data and output directories



1. **geog_data_path**: your WRF_GEOG folder.
2. **Geogrid output directory**: where the geo_em files are written.
3. **Ungrib output directory**: where the intermediate files are written.
4. **Ungrib file prefix**: for example FILE. The namelist gets the directory and prefix joined, such as /home/user/wrf_output/ungrib/FILE.
5. **Metgrid output directory**: where the met_em files are written.
6. **Browse buttons**: pick a folder. Browsers hide full paths, so only the folder name replaces the last part of the existing path. Check the start of the path afterwards.

Step 7. Generate and export the namelists

The message under the domains table confirms success, or lists warnings if a nest does not fit inside its parent.



1. **Generate namelists**: builds namelist.wps and namelist.input from your domains and settings.
2. **Save project**: saves domains, settings and paths as a .json file.

3. **Editors**: read and edit the generated text before using it.
4. **Copy and Download (namelist.wps)**: copy the text to the clipboard or save the file.
5. **Copy and Download (namelist.input)**: the same for the WRF input namelist.
6. **Open project**: load a saved .json file to continue later.

Checking before a real run

- The grid sizes and parent start positions are estimated from the boxes you draw. Run geogrid.exe and plot the result to confirm the domains.
- e_we - 1 and e_sn - 1 of each nest must divide by its parent ratio. The tool aligns them, and warns about problems.
- The physics and dynamics values are general defaults. Adjust them for your case.

Troubleshooting

Problem	What to do
Map is blank	Check your internet connection, then hard-refresh with Ctrl+Shift+R.
Cannot drag the map	Untick Draw mode.
Cannot move or resize a box	Untick Draw mode and select the domain first.
"Cannot generate" message	Make sure every domain has numeric extents and a resolution, and that End is after Start.
Browse shows an upload prompt	Your browser lacks the folder picker. Nothing is uploaded; you can type the path instead.